

Mode of Examination: Online
M.Sc. Semester- II Examination, 2021

2021

Subject: Computer Science

Paper Code & Name: CSM 205(P) Object Oriented Programming(Module 1)

Full Marks: 20

Date: 8/19/2021

Time: 1:30 hours

Duration:

11.00 am to 12.30 pm

Please follow the instructions below carefully.

Promise not to commit any academic dishonesty

Candidates are required to answer in their own words as far as applicable.

Each Page of answer scripts should have your examination Roll Number on the right side top corner of your answer script / Script may be a softcopy also.

The name of the scanned copy of the assessment script will be of the following format:

CSM203-CN-Roll-Number.pdf

(Example: CSM203-C91-CSC-201001.PDF)

The subject of the mail should be the file name only.

The scanned answer script is to be sent to tocucse2020@gmail.com

Extra credit for the format of document & presentation

The report should have an index page.

30 minutes is allotted for scanning and uploading your answer script.

The answer script may not be accepted after the scheduled time.

Answer Any One Question

According to the Given List bellow where the Roll Number is mapped with a Question No

Set 1

- A. Create a class Parent having instance variables id, name and address. Create a class ChildOne having instance variables id, name, address and marks. Also create another class ChildTwo with instance variables id, name, address, qualification and salary. Within each class define your own method to display values of these variables. Design the program using super call with proper parameter and use object of each class from main() to display their properties.
- B. Write a program to show addition, subtraction and multiplication of two matrices.

Set 2

- A. Write a program to find the factorial of a given integer number using recursion (take input using command-line argument).
- B. Assume that a bank maintains two kinds of account for its customers, one called savings account and other called current account. The savings account provides compound interest and withdrawal facilities but no cheque book facility. The current account provides cheque book facility but no interest. Current account holders should also maintain a minimum balance (say Rs. 1000) and if the balance falls below this level a service charge is imposed (say Rs. 100). Create a class **Account** that stores customer name, account number and type of account. From this class derive two classes **Curr_Acct** and **Savn_Acct** respectively to make them more specific to their requirements. Include the necessary methods to achieve the following tasks:
 - a. Accept deposit from a customer and update the balance.
 - b. Display the balance.
 - c. Compute and deposit interest.

- d. Permit withdrawal and update the balance.
- e. Check for minimum balance, impose penalty, if necessary, and update balance.
- f. Use constructors to initialize the class members.

Set 3

- A. Overload the constructors for classes Area and Volume of a rectangular figure and also display its area and volume. Area is the superclass and Volume is the subclass.
- B. Write a program to do the following:
 - a. To output the question “Who is the inventor of Java”?
 - b. To accept an answer.
 - c. To print out “Good” and then stop, if the answer is correct.
 - d. To output the message “Try Again” and then stop, if the answer is wrong.
 - e. To display the correct answer when the answer is wrong even at the third attempts and stop.

Set 4

- a) Generate password from initials of one’s first_name, middle_name, last_name and with last four digit of your roll_no (if middle name is not present, it won't come).
- b) Create a class **Employee** is having instance variables name and id. Create its subclass named **Scientist** which has instance variables no_of_publication and experience. Now create its subclass, say **DScientist** which has an instance variable award. Put a method like: **public String toString(){ }** in every class where you describe about the class and from main() method create object of each class and print each object.

Set 5

- a) WAP to calculate GCD of two numbers.
- b) Create a class Parent having instance variables id, name and address. Create a class ChildOne having instance variables id, name, address and marks. Also create another class ChildTwo with instance variables id, name, address, qualification and salary. Within each class define your own method to display values of these variables. Design the program using super call with proper parameter and use object of each class from main() to display their properties.

Set 6

- a) Create a class with a method void show() and make three subclasses of it and all subclasses have this show() method overridden and call those methods using their corresponding object references.
- b) Write a program in Java that accepts a shopping list of five items from the command line and stores them in a vector. Now modify it to:
 - To delete an item in the list.
 - To add an item at a specified location in the list.
 - To add an item at the end of the list.
 - To print the contents of the vector.

Set 7

- a) Write a program to find the sum and average of several integers (in an array) using enhanced-for loop.
- b) Write a program in Java to create a StringBuffer object with content "Object Programming languages". Then perform the following operations:
 - Print the object and determine its length and capacity.
 - Insert a new string "Oriented" after the word "Object". Print the modified string.
 - Insert a character '-' at the 6-th position. Print the modified string again.
 - Append another string named " are very popular." Print the resulting string.
 - Now delete the character '-' and put a space (" ") there. Print it.
 - Apply these operations one-by-one: (a) delete(1, 7), (b) delete(2, 10), and (c) delete(3, 13).

Then print the result.

- Convert the StringBuffer type object into a String object. Print the final result.

Set 8

Create a Student class with proper attributes like roll no, name, stream, semester and college.

- a) From main create such three students and show their information.

- b) Write a JAVA Program considering the above mentioned Student class.

Assume that students study varying number of subjects. Each subject has a title, internal marks and theory marks. Write a Program to define Student class including the subjects as **vararg** argument of constructor. From main create such three students and show their information including subjects' name and grand total marks.

Set 9

- a) Create an interface named Shape with a field pie (=3.14). Create two subclasses of it named Circle and Rectangle. Create object of the two classes and calculate their area.
- b) Write a program that takes one string and two integers as command line argument and prints the reverse of the substring of the string specified by the two numbers. The program should handle all possible exception that may arise due to bad input.

Set 10

- a) Create a class with two methods, f() and g(). In g(), throw an exception of a new type that you define. In f(), call g(), catch its exception and, in the catch clause, throw a different exception (of a second type that you define). Test your code in main() method.
- b) Write a program that takes one string and two integers as command line argument and prints the reverse of the substring of the string specified by the two numbers. The program should handle all possible exception that may arise due to bad input.

